1. KVL: $32V - V_1 - 8V + V_2 = 0$

$$V_1 - V_2 = 24V$$

$$i_1 = i_2 \Rightarrow V_1 = 4\Omega \cdot i_1 = 4\Omega \cdot i_2$$

$$= -\frac{4\Omega}{2\Omega} \cdot V_2 = -2V_2$$

$$-3V_2 = 24V \Rightarrow V_2 = 8V$$

$$V_1 = -2V_2 = 16V$$

2. KVL: $70V - V_x - 2V_x + V_0 = 0$

$$3V_x - V_0 = 70V$$

$V_x = 10\Omega \cdot i_1$ $\qquad V_x = 2V_0 \quad (i_1 = i_2)$

$-V_0 = 5\Omega \cdot i_2$ $\qquad -7V_0 = 70V$

$$V_0 = -10V$$

$$V_x = -2V_0 = 20V$$

3. $$R_{eq} = 4\Omega + \frac{1}{\frac{1}{6\Omega} + \frac{1}{4\Omega + \frac{1}{\frac{1}{12\Omega} + \frac{1}{8\Omega + 6\Omega + 10\Omega}}}} = 8\Omega$$

4. $$\frac{1}{C_{eq}} = \frac{1nF^{-1}}{\frac{1}{50} + \frac{1}{70}} + \frac{1nF^{-1}}{\frac{1}{20} + \frac{1}{\frac{1}{60} + \frac{1}{20+10}}} = \frac{91}{3720} nF^{-1}$$

$$C_{eq} = \frac{3720}{91} nF \approx 40.88 nF$$

5. KVL: $150V - V_1 - V_2 = 0$

$$V_2 - V_3 - V_4 = 0$$

$$C_{eq} = \frac{1}{\frac{1}{40} + \frac{1}{\frac{1}{20} + \frac{1}{\frac{1}{60} + \frac{1}{30}}}} = 20 \mu F$$

$$q = C_{eq} \cdot V = 20 \cdot 10^{-6} \times 150C$$

$$= 3 \times 10^{-3} C$$

$$V_1 = \frac{q}{C_1} = \frac{3 \times 10^{-3}}{4 \times 10^{-5}} = 75V$$

$$V_2 = 150V - V_1 = 75V$$

$$C_{3,4} = \frac{1\,\mu F}{\frac{1}{60} + \frac{1}{30}} = 20\,\mu F$$

$$V_3 = \frac{V_2 \cdot C_{3,4}}{C_3} = 25V$$

$$V_4 = V_2 - V_3 = 50V$$

$V_1 = 75V$

$V_2 = 75V$

$V_3 = 25V$

$V_4 = 50V$

6. $$L_{eq} = 4\,H + \left(\frac{1}{\frac{1}{7} + \frac{1}{20 + \frac{1}{12 + 10}}}\right) H + 8\,H = 18\,H$$

7. $$L_{eq} = \frac{1}{\frac{1}{50} + \frac{1}{20 + \frac{1}{\frac{1}{60} + \frac{1}{100 + \frac{1}{\frac{1}{30} + \frac{1}{40 + 20}}}}}} = 25\,mH.$$